Electronic Devices and Sensors

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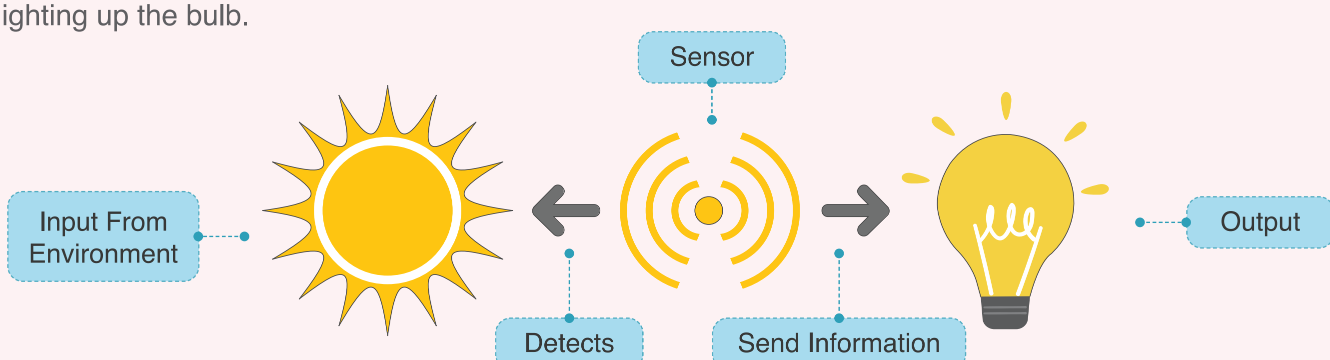
What is a sensor?

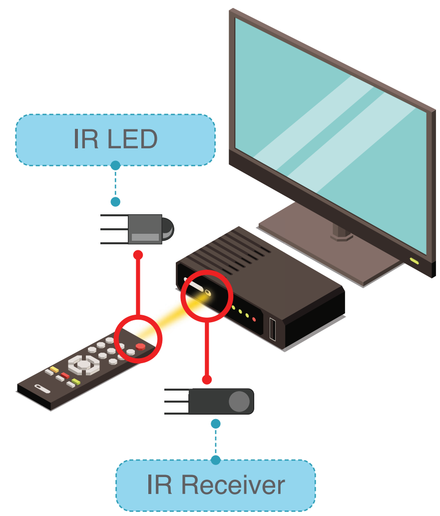
A sensor is an electronic device which detects and responds to input signals from the environment. Different types of sensors are used in our offices, gardens, shopping malls, homes, cars, toys etc. They make our lives easy and comfortable. Home appliances like fans, fire alarms, TVs, air conditioners, reverse cameras on cars, and digital thumb impressions all use sensors.

For example: In the image below, a sensor receives solar energy as an input signal and responds to it by lighting up the bulb.



Let's look at how sensors communicate with other devices.

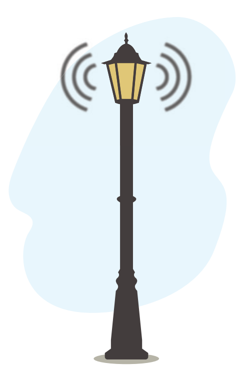
We use remotes to control a television or an AC. Most remotes use an IR sensor that emits an infrared signal. The television or AC responds to these signals.



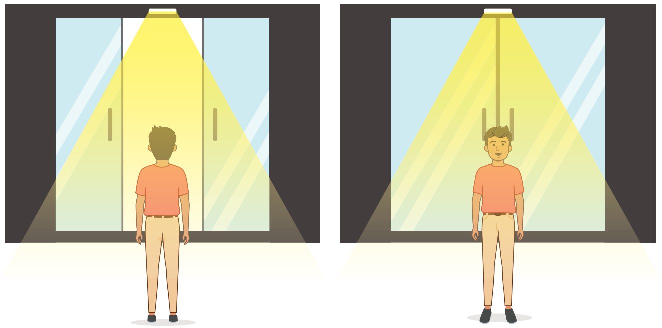
As per the input, sensors communicate with other devices in the same way.

What are some everyday objects that use sensors?

a. Some streetlights use sensors to detect movement and turn on.



b. Doors that automatically open when a motion sensor detects a person coming close.



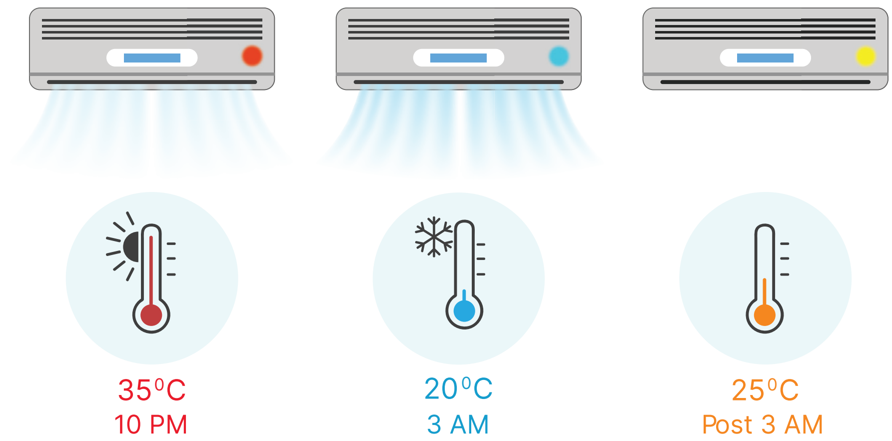
Bytes

Transducer: A device that converts energy from one form to another.

Sensor: A sensor can convert energy in the form of light, temperature, motion, moisture, pressure, etc. into electrical energy.

Actuators: A device which converts electrical energy into physical output. Actuators act as output devices. For example: Buzzer and LED lights.

c. A temperature sensor in cooling or heating devices which detects the surrounding temperature.

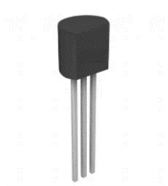


Sensors and their applications in daily life

a. Temperature sensor

A temperature sensor is a device that measures the temperature of environment.

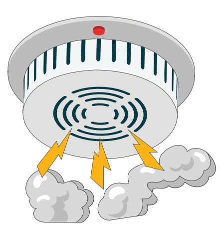
For example: Temperature sensors are used in refrigerators and freezers to maintain an optimum temperature.



b. Smoke sensor

A smoke sensor is an electronic fire-protection device that automatically senses the presence of smoke during a fire.

For example: smoke sensors are used in smoke detectors in malls, theatres, and apartments.



Smoke sensor application

c. Ultrasonic sensor

An ultrasonic sensor is used to measure the distance to an object using ultrasonic sound waves.

For example: Cars use an ultrasonic sensor for parking assistance.

Ultrasonic sensor application

Activity

Create an automatic light control circuit to understand the working of a sensor.

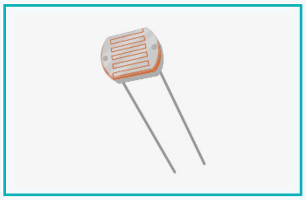
Automatic Light Control Circuit

In an automatic light control circuit, the LED turns on or off to indicate the light intensity, or the presence or absence of light in that area.

Light in home appliances is usually controlled manually. This may lead to power wastage due to human error or unusual circumstances. To avoid such circumstances, an automatic light control circuit is used.

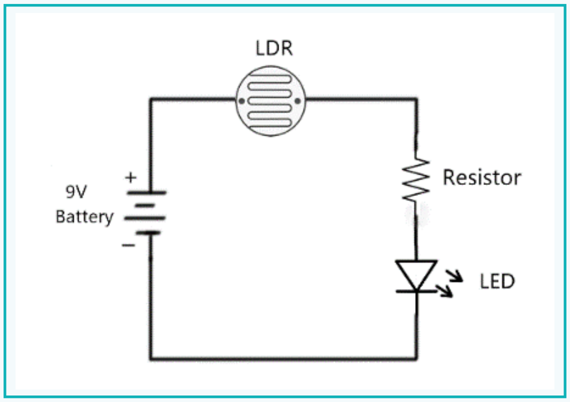
Working: The circuit uses an LDR sensor, due to which the LED will go on and off as per the available light.

LDR sensor: LDR stands for Light Dependent Sensor, also called a photo resistor. An LDR sensor is a light sensitive device that indicates the presence or absence of light. In the presence of light, the resistance of the LDR decreases and current starts flowing through it. In absence of light, it provides more resistance.



LDR sensor

Circuit Diagram:



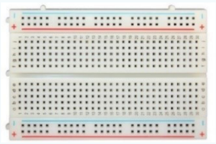
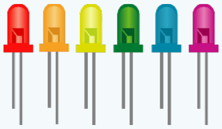
Automatic light control circuit

If the available light is low in the room or in the surroundings, then the LED will be OFF. If the available light is greater, then the LED will be ON.

This functionality can be reversed and used in automated light control applications.

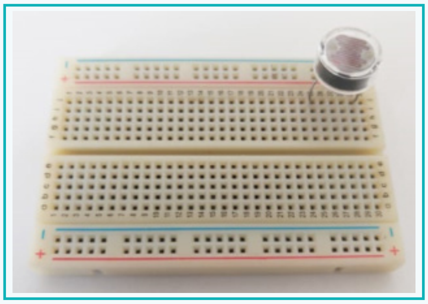
Activity

Required components and quantities for a temperature circuit:

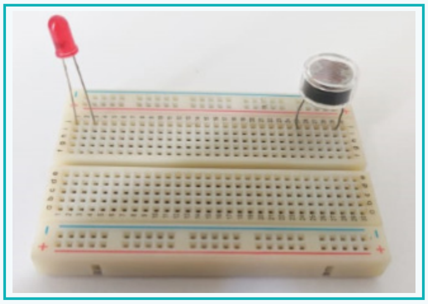
Sr. No.	Component Name (Required component)	Image of component	Quantity
1	Breadboard		1
2	LDR SENSOR (Light Dependent Resistor)		1
3	LDR		1
4	Resistor 10 K ohms		1
5	9V Battery		1
6	Jumper Wires		

Activity

1. Take the breadboard and place the LDR SENSOR on it.

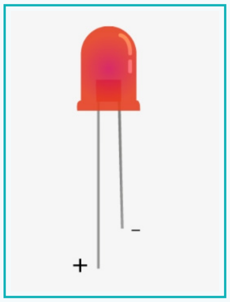


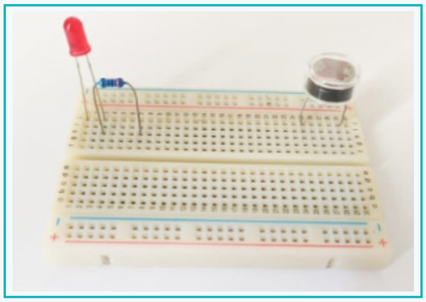
2. Place an LED on the breadboard (the LED can be of any colour).



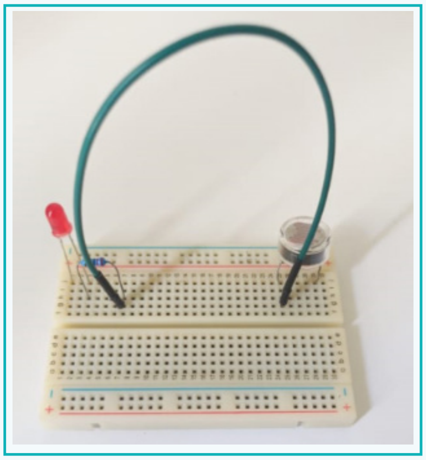
3. Connect a 220-ohm resistor to the positive terminal of the LED. The resistor limits the current passing through the LED.

Note:
An LED's longer leg is positive and the shorter one is negative.



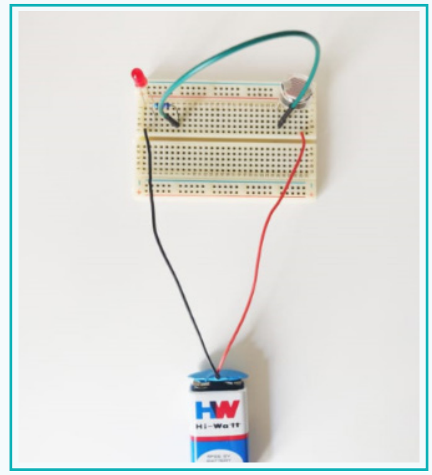


4. Use a jumper wire to connect another resistor terminal to a terminal of the LDR sensor.

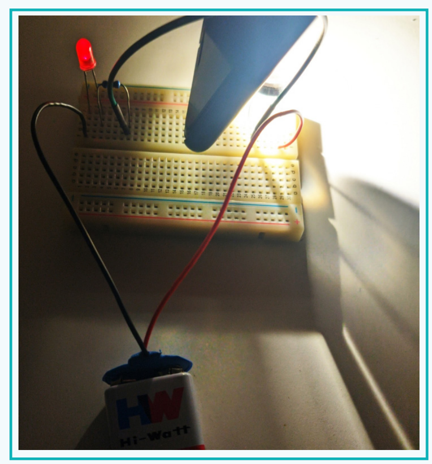


Activity

5. Connect the resistor's negative terminal to the 9V battery's negative terminal. Connect the battery's positive terminal to the LDR sensor.

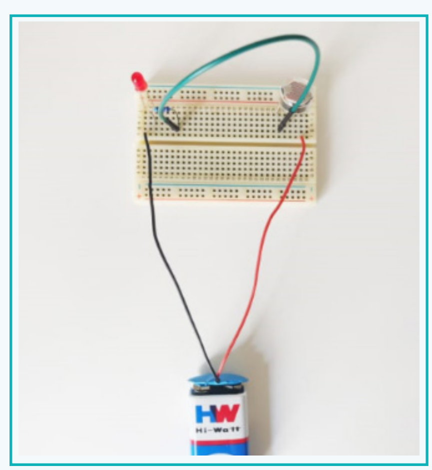


6. Now move the LDR sensor into a bright area to see changes in the LED.



LED in dark area, LED ON

7. Move the LDR sensor into a dark area again to observe the change.



LED in a bright area, LED OFF

This lesson took us through an overview of sensors and electronic devices. A sensor is a device that receives a signal or stimulus and responds in the form of an electrical signal. Its output signals correspond to some form of electrical signal, such as current or voltage. A sensor receives different kinds of signals i.e. physical, chemical, or biological, and converts them into an electric signal. We looked at everyday objects which use sensors and completed an activity to create an automatic light control circuit.

Exercise

Fill in the blanks

1. Sensors are used to detect and collect information about the respective events or changes in ________________.
2. An LDR sensor is used to indicate the presence or absence of ________________.
3. If the LDR is in a dark area, then the LED will turn ________________.
4. The longer leg of an LED is ________________.
5. The shorter leg of an LED is ________________.

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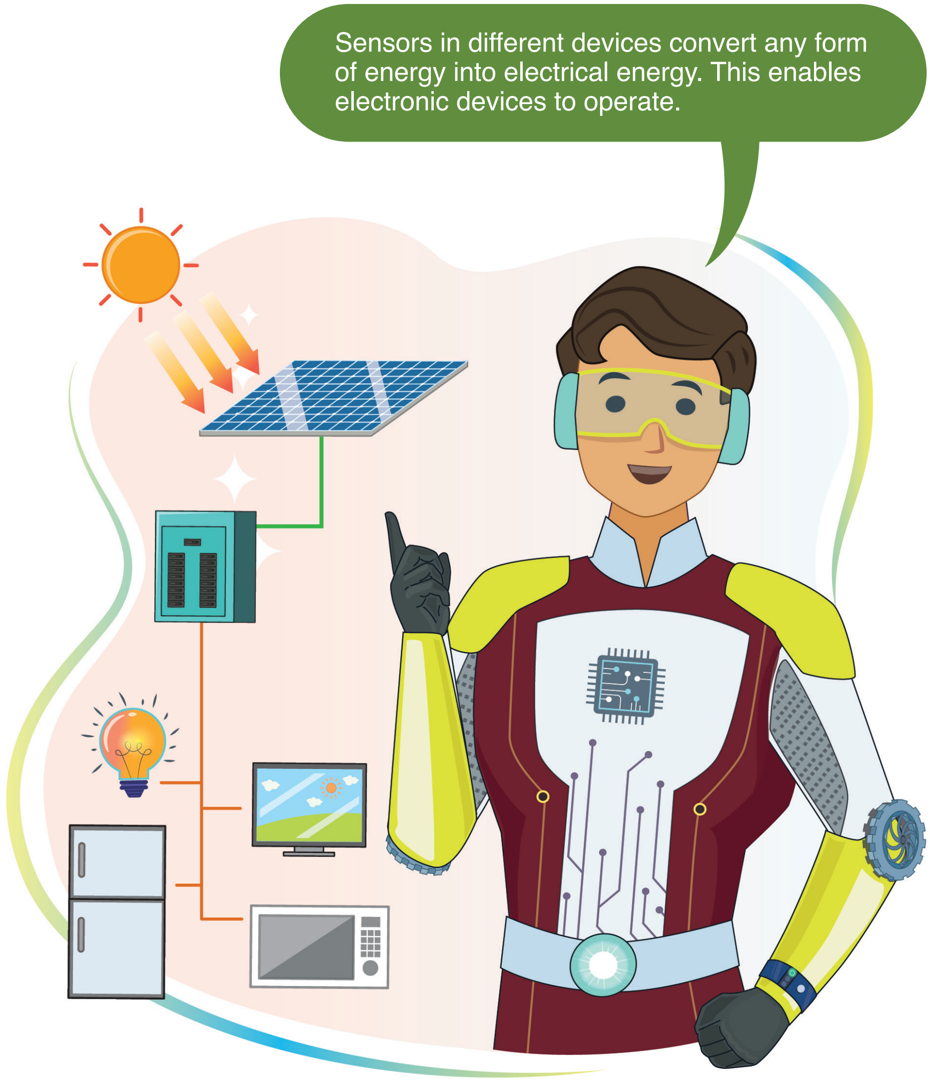

Tech Flash

- **Sensor** : A device which responds to environmental input.
- **Breadboard** : A board used to hold and connect electronic components.
- **LDR sensor** : A light-dependent sensor used to indicate the presence or absence of light.